Quantum Isomorphisms of Finite Graphs – Two Strongly Regular Graphs from the Penrose Dodecahedron

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Abstract. In this report we study a relation between graphs known as quantum isomorphism which come from the study of non-local games in quantum information and naturally generalize the notion of a graph isomorphism. We give an overview of the few known examples of non-isomorphic quantum isomorphic graphs and aim to extend this list by analysing the orthogonality graph Q of the Penrose dodecahedron and its clique graph Q^*, both of which are strongly regular with parameters $(40, 12, 2, 4)$ and have surprisingly many properties in common. While we were neither able to prove nor to disprove the existence of a quantum isomorphism between Q and Q^*, we report several observations which strengthen our hypothesis and show how both graphs may be understood in terms of the 2-qutrit Pauli group.

Keywords: Quantum isomorphism · Non-local games

1 Introduction

The recent advances in quantum information and quantum computation have profoundly transformed computational science. There are many scientific fields that will potentially benefit from the existence of a full-scale quantum computer, including economics, physics (via efficient simulation of quantum systems) and even machine learning (see e.g. [8]). However, the most significant and impactful application of quantum computing is in the field of cybersecurity, as Shor's algorithm [21] allows to break conventional asymmetric cryptography based on prime factorization or discrete logs exponentially faster than any classical computer. This discovery fundamentally disrupted cryptographic research, paving the way for the emergence of a new field called post-quantum cryptography, in which new cryptographic systems are designed which conjecturally resist the dangers of quantum algorithms. Moreover, quantum phenomena such as superdense coding [10] and quantum teleportation [4] lays the groundwork for the field of quantum cryptography, which is the study of quantum protocols resistant to eavesdropping.

A formal method to quantify these seemingly information-theoretic advantages of quantum physics comes from the study of *non-local games*. A non-local

game is an interactive cooperative game, in which the players win by fulfillling a given relation between the questions that are given to each player and their respective answers. While there is no communication allowed between the players during the game, possibly entangled quantum resources are allowed to be shared. By leveraging this quantum advantage it is possible for the players to achieve winning probabilities that exceed those of any classical (i.e. non-quantum) strategy.

One of the most well-known non-local game is the magic square game introduced in [2], in which a 3×3 grid of ± 1 entries must be (partially) filled by the players, such that each row multiplies to 1 and each column multiplies to -1. The first player gets asked the assignments of a specific row, while the second player is asked about a column. The game is won if the mutual entry of the magic square is filled with the same values by both players. While any strategy that involves entirely filling the magic square beforehand and answering according to this assignment is impossible by design, it is possible to leverage the quantum resources to win the game with certainty. Due to this seemingly magical phenomenon, the magic square game is an example of what is called a *pseudo-telepathic* game.

Another example of such a *pseudo-telepathic* game is the graph isomorphism game, which is parametrized by two publicly known graphs G and H. The referee's questions to each player comprises of a vertex of G, to which the players must answer with a vertex of H. In order to win, the nodes given as answers must fulfill the same adjacency relations in H as the nodes that are given as questions by the referee do in G. While one can construct a perfect classical game strategy if and only if G and H are isomorphic, it is known that non-isomorphic graph pairs exist for which a perfect quantum strategy can be found. Such graph pairs are said to be quantum isomorphic.

Contribution. We analyse a new pair of strongly regular graphs based on the Penrose dodecahedron and show that the properties of those graphs suggest that they are quantum isomorphic. While we neither succeeded to prove nor to disprove this hypothesis, we discovered that Q has an quantum independence number of at least 10, which is larger than its classical independence number and matches the independence number of Q^*, fitting into the scenario of the graphs being quantum isomorphic. Moreover, we provide a natural description of Q as the commutation graph of the elements inside the 2-qutrit Pauli group, while Q^* gives the graph of maximally commuting subsets of that group.

Structure of this work. We introduce the basic notions relevant for this report in Section 2, which includes basics from graph theory, and a brief introduction to non-local games, particularily to the graph isomorphism game, which is the central game studied in this report. Next, we give a survey on known examples for quantum isomorphic graphs in Section 3, which we aim to extend in Section 4 by analysing the Penrose dodecahedron. The observations in Section 4 are com-

plemented by the discussion in Section 5, which describe structural obstructions when searching for a magic unitary via the method from [20].

2 Preliminaries

2.1 Graph Theory Basics

In this report, we consider the structure of finite graphs. Therefore, we go by the convention that a graph $G = (V(G), E(G))$ comprises of a finite set of vertices $V(G)$ and a set of undirected edges $E(G) \subseteq V(G) \times V(G)$, meaning that $(i, j) \in E(G)$ iff $(j, i) \in E(G)$. Throughout, we also use the notation $i \sim_G j$ to mean $(i, j) \in E(G)$. Furthermore, we omit the graph subscript for this notation, whenever there is no ambiguity.

An independent set of a graph is a subgraph of G containing no edges, and the independence number $\alpha(G)$ is defined to be the vertex count of a maximal independent set of G.

A homomorphism of two graphs G, H is defined to be a map $\varphi : V(G) \to V(H)$ such that $i \sim_G j \implies \varphi(i) \sim_H \varphi(j)$. Moreover, two graphs G and H are isomorphic if the above implication is two-sided. An automorphism of a graph G is an isomorphism to itself.

A strongly regular graph with with parameters (n, k, λ, μ) is a graph G with the properties

- $|V(G)| = n$,
- Each vertex has exactly k neighbours,
- Any two adjacent vertices share exactly λ neighbours,
- Any two non-adjacent vertices share exactly μ neighbours.

Strongly regular graphs have been one central object of study in algebraic graph theory, see for example [7], as they lie on the border between structure and disorder and are still poorly understood in general. In particular, strongly regular graphs are generally not uniquely determined by their parameters, and for some parameter choices there are exponentially many non-isomorphic graphs, while for other parameter choices there are none at all.

2.2 Non-Local Games

Non-local games consist of two non-communicating players Alice and Bob, which are given questions $x \in I_A, y \in I_B$ by the referee, to which they must respond with an answer $a \in O_A, b \in O_B$. The game is won if the questions and answers fulfill some predefined condition, captured by a *verification function*

$$V : I_A \times I_B \times O_A \times O_B \ \to \ \{0, 1\},$$

i.e., if $V(x, y, a, b) = 1$, the game is won. The probability for the players to receive questions x and y are modelled via some probability distribution $\pi : I_A \times I_B \to [0, 1]$, which, for simplicity, is generally assumed to be a uniform distribution.

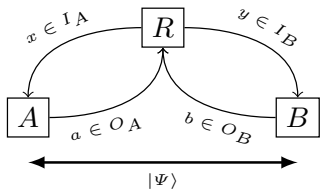


Fig. 1. Graphical illustration of how non-local games are played.

The caveat to non-local games is that the players are not allowed to communicate during the game, but are allowed to use entangled *quantum* resources which are distributed before the game.

We therefore distinguish between two types of strategies, namely classical and quantum strategies. We say that Alice and Bob adopt a (deterministic) *classical strategy* if they agree on functions $f_A : I_A \to O_A$ and $f_B : I_B \to O_B$ beforehand, which are applied to their questions during the game. In particular, no quantum entanglement is used. On the other hand, in case of a *quantum strategy*, the players agree to apply local question-dependent operations, i.e. unitaries and projection-valued measures (PVMs), onto their share of qudits, and answer depending on the result of some measurement on their share of qudits.

The *winning probability* of a non-local game G for some strategy S is defined as

$$\omega_S(G) = \sum_{x,y} \pi(x,y) \sum_{a,b} V(x,y,a,b)\ p(a,b|x,y)$$

where $p(a,b|x,y)$ is the probability for Alice and Bob to respond with a, b, given questions x, y. Note that this is the only term that the players have control over, namely by agreeing on a game strategy S beforehand, while the other terms are predetermined by the game instance that is played. In particular, for classical strategies, we may simplify to

$$p(a,b|x,y) = p\left(f_A(x) = a\right) \cdot p\left(f_B(y) = b\right),$$

while for quantum strategies we may rewrite to

$$p(a,b|x,y) = \langle \Psi | P_x(a) \otimes P_y(b) | \Psi \rangle,$$

where P_x and P_y are PVMs parametrized by the received questions. Studying non-local games is motivated by a peculiar phenomenon called *quantum advantage*, describing that the pre-shared quantum resources allow to devise quantum strategies, which may achieve higher winning probabilities than classical strategies, i.e. strategies that do not utilise these quantum resources. Phrased in terms of the optimal classical and quantum winning strategies,

$$\omega^*(G) := \sup_{S \text{ classical}} \omega_S(G), \qquad \omega_q^*(G) := \sup_{S \text{ quantum}} \omega_S(G),$$

there exist non-local games G for which $\omega^*(G) < \omega_q^*(G)$. Furthermore, it is known that quantum strategies are non-signalling, meaning that the answers of the players are statistically independent of each other, defying intuition that during the game, some form of information is exchanged via those local operations on the shared entangled pair.

The Mermin-Peres Magic Square Game. A prominent example of a non-local game was introduced by Mermin and Peres [14] and is based on assigning values to a 3 by 3 square.

Definition 1. *The* magic square game *is a non-local game based on a 3 by 3 grid with entries $x_i \in \{1, -1\}$, such that the product of the x_i over one row must be 1, while the product over one column must be -1.*

The non-local game is then defined as follows. The referee asks Alice to fill in some row and Bob to fill in some column. The players win if their assignments are valid and if both players assign the same value to the element contained in the intersection.

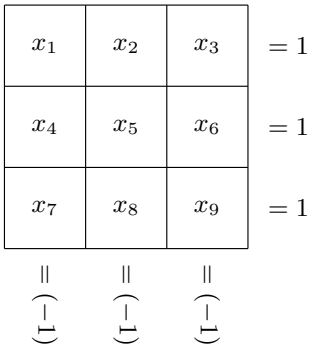


Fig. 2. The Magic square game.

Note that it is impossible to find a classical strategy with winning probability one, as it would imply filling out the entire magic square, which is impossible by design. To see this, compare the product of all rows to the product of all columns. On the one hand, they both compute the product of all entries $\prod_{i=1}^{9} x_i$ and should therefore be equal. On the other hand, $1 \cdot 1 \cdot 1 = 1 \neq -1 = (-1) \cdot (-1) \cdot (-1)$, giving us the contradiction.

However, it is in fact possible to find an optimal quantum strategy, by sharing two qubit pairs in the Bell state, i.e. the state

$$|\Psi\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle|0\rangle + |1\rangle|1\rangle\right)_{A_1 B_1} \otimes \frac{1}{\sqrt{2}} \left(|0\rangle|0\rangle + |1\rangle|1\rangle\right)_{A_2 B_2},$$

where $|0\rangle = (1,0)^\mathsf{T}$ and $|1\rangle = (0,1)^\mathsf{T}$ are the standard basis vectors of the Hilbert space $\mathbb{C}^2$, in which the possible states of a qubit live. Now, based on the questions they received, Alice and Bob measure their share of qubits via the observables given in Figure 3 and label the square entries with the measurement outcomes, i.e. with one of the eigenvalues of the observable. Since the operators in one row or column commute with each other, they share an eigenbasis, and since quantum measurements translate to projections onto eigenspaces of the observable, the order of measurement does not affect the outcomes. Furthermore, as Alice and Bob share two copies of the Bell state, it means that measuring with the same observable gives the same outcomes for both players, therefore their answers on the intersecting entry always matches. Moreover, the product of the eigenvalues inside one row or column is equal to the eigenvalue of their product, which is by design $I \otimes I$ for the rows and $-I \otimes I$ for the columns, with eigenvalues 1 and -1 respectively. Therefore, the assignment produced by the quantum strategy is valid. Lastly, since no probabilities are involved, we get a quantum strategy with a winning probability of one.

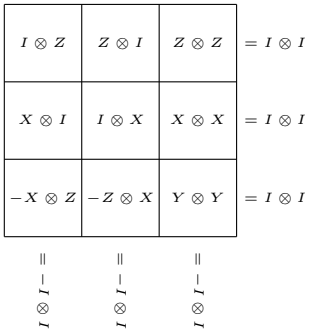


Fig. 3. An optimal quantum strategy to the magic square game.

The Graph Isomorphism Game. We define another non-local game, which is based on graph isomorphisms.

Definition 2. *Let G, H be graphs with the same number of vertices. The (G,H)-isomorphism game (or graph isomorphism game) is the non-local game defined by the following rules:*

- *Alice and Bob both receive vertices $v_A, v_B \in V(G)$*
- *They have to answer with vertices $w_A, w_B \in V(H)$*
- *The game is won if $rel_G(v_A, v_B) = rel_H(w_A, w_B)$, where*

$$rel_\Gamma : \ V(\Gamma) \times V(\Gamma) \ \rightarrow \ \{=, \sim, \not\sim\}$$

denotes the adjacency relation between two vertices of some graph Γ.

Based on this definition, we further define the notion of a quantum isomorphism between graphs.

Definition 3. *Two graphs G, H with the same number of vertices are said to be* quantum isomorphic*, denoted $G \cong_q H$, if there exists a perfect quantum strategy for the (G, H)-isomorphism game.*

Remark 1. An immediate observation from those definitions is that the game has an optimal classical strategy when the graphs G and H are isomorphic, as applying this isomorphism gives such a strategy. Since the quantum strategy of simply ignoring the quantum resources and employing this classical strategy is already optimal, we have

$$G \cong H \implies G \cong_q H.$$

Interestingly, the converse is not true. As we will see repeatedly, there exist pairs of graphs which are quantum isomorphic yet non-isomorphic. These pairs of graphs are the main object of study in this work and to this date, very few examples of these are known. Throughout this work, we will simply call such graph pairs *quantum isomorphic* and implictly assume that they are non-isomorphic, unless stated otherwise.

There are many alternative definitions of a quantum isomorphism between graphs, some of which are more helpful for the explicit construction of those graph pairs. Below we give one such definition, which is stated in more graph theoretic terms.

Definition 4 ([23]). *Let $\mathcal{A} \cong M_m(\mathbb{C})$ for some positive integer m. A* magic unitary *(or quantum permutation matrix) is a matrix $u \in M_n(\mathcal{A})$ whose entries are projections in $\mathcal{A}$, i.e. $\forall i, j \in \{1, \dots, n\} : \ u_{ij} = u_{ij}^* = u_{ij}^2$, and furthermore*

$$\forall i \in \{1, \dots, n\} : \ \sum_k u_{ik} = \sum_k u_{ki} = \mathbf{1}_{\mathcal{A}} \tag{1}$$

Remark 2. Note that this definition generalizes the notion of a permutation matrix, since the definitions coincide for the special case $\mathcal{A} = \mathbb{C}$ (here the set of projections is just $\{0, 1\}$).

Lemma 1. *Let G, H be graphs. G and H are quantum isomorphic iff there exists $\mathcal{A} \cong M_m(\mathbb{C})$ for some positive integer m and a magic unitary $u \in M_n(\mathcal{A})$ such that*

$$(A_G \otimes I_m)u = u(A_H \otimes I_m),$$

where A_G and A_H are the adjacency matrices of G and H respectively.

Remark 3. Continuing the previous remark, we see that the special case $\mathcal{A} = \mathbb{C}$ translates to quantum isomorphisms being equivalent to classical isomorphisms, as the adjacency matrices of classically isomorphic graphs are conjugate under some permutation, represented by a permutation matrix.

The condition that u interchanges the adjacency matrices of G and H may be reformulated in terms of products of the entries of u, which gives a more concrete approach to showing the quantum isomorphism induced by u.

Lemma 2 ([20]). *G and H are quantum isomorphic if there exists a magic unitary $u \in M_n(\mathcal{A})$, for which $u_{ks}u_{lt} = 0$ for $(k,l) \in E(G)$ and $(s,t) \notin E(H)$, or vice versa.*

The (G,t)-independent set game ([13]). Continuing with non-local games related to graphs, we also define one which relates to the graph property of containing a t-independent set.

Definition 5. *Let t be a natural number and G be a graph. Then, the* (G,t)-independent set game *is defined as follows.*

- *Alice and Bob receive natural numbers $1 \leq a, b \leq t$*
- *They answer with vertices $x_a, x_b \in V(G)$*
- *The game is won if $a = b \implies x_a = x_b$ and $a \neq b \implies x_a \not\sim x_b$.*

More generally, we define the notion of a quantum independent set of a graph G, as projections that yield an optimal quantum strategy for the (G,t)-independent set game.

Definition 6. *Let G be a graph and $\mathcal{A} \cong M_m(\mathbb{C})$ for some positive integer m. A* quantum independent set *of size t is a collection $P = \{P_{xu}\}_{1 \leq x \leq t, u \in V(G)}$ of projections $P_{xu} \in \mathcal{A}$ such that*

- *$\sum_{u \in V(G)} P_{xu} = \mathbf{1}_{\mathcal{A}}$ for all $1 \leq x \leq t$*
- *$P_{xu}P_{yv} = 0$ for $(u,v) \in E(G)$ and all $1 \leq x, y \leq t$*
- *$P_{xu}P_{yu} = 0$ for all $x \neq y, u \in V(G)$*

Furthermore, the quantum independence number *$\alpha_q(G)$ is defined as the maximal size of a quantum independent set of G.*

By construction, the game has an optimal classical strategy if G contains an independent set of size t, as the strategy then involves labelling the independent set as $x_1, \ldots, x_t$, giving an optimal strategy. However, as shown in [13], optimal quantum strategies do not imply the existence of a t-independent set in G, meaning that there exist graphs G for which the classical independence number $\alpha(G)$ is strictly smaller than $\alpha_q(G)$.

3 Known Examples of Quantum Isomorphic Graphs

In this section we give a survey on known examples of quantum isomorphic graphs. We start by showing that the magic square game gives quantum isomorphic pairs of graphs and continue with more a general construction, which does not involve binary constraint games.

3.1 Quantum Isomorphic Graphs from the Mermin-Peres Magic Square Game

It is well-known that one can construct graphs $G_{\mathcal{F}}$ and $G_{\mathcal{F}_0}$ based on any linear binary constraint system game $\mathcal{F}$, such that the optimal quantum strategy directly translates to a graph isomorphism between $G_{\mathcal{F}}$ and $G_{\mathcal{F}_0}$, as demonstrated in [3]. In particular, we summarize this reduction for the magic square game, which falls under this class of non-local games.

First, note that each constraint coming determining the product of entries over a row or column, say $x_1x_2x_3 = 1$, can be translated to constraints modulo two on the exponents, namely $z_1 \oplus z_2 \oplus z_3 = 0$, when we write $x_i = (-1)^{z_i}$. Considering those exponents, we construct a graph $G_{\mathcal{F}}$ with vertices (ℓ, f) coming from valid assignments f for those additive constraints C_ℓ. For example, $(z_1 \oplus z_2 \oplus z_3 = 0, \mathbf{101}) \in V(G_{\mathcal{F}})$ while $(z_1 \oplus z_2 \oplus z_3 = 0, \mathbf{001}) \notin V(G_{\mathcal{F}})$. Next, two vertices in $G_{\mathcal{F}}$ are adjacent when the assignments are inconsistent, meaning that the assignment of the shared value is different. For example,

$$((z_1 \oplus z_2 \oplus z_3 = 0, \mathbf{101}), (z_1 \oplus z_4 \oplus z_7 = 1, \mathbf{010})) \in E(G)$$

as the shared value z_1 is inconsistent.

For the other graph $G_{\mathcal{F}_0}$, we consider a variant $\mathcal{F}_0$ of the magic square game, which is obtained by setting the right-hand side of all constraints ℓ to zero. The adjacency rules for $G_{\mathcal{F}_0}$ remain the same as for $G_{\mathcal{F}}$. Note that by construction, the graph $G_{\mathcal{F}_0}$ has an independent set of size six, as the all-zero assignment is valid and consistent for any row and column, which means that the corresponding six vertices are all non-adjacent to each other. The statement that $G_{\mathcal{F}}$ and $G_{\mathcal{F}_0}$ are quantum isomorphic but non-isomorphic follows from the two theorems below, which state the reductions from optimal classical strategies to classical isomorphisms and optimal quantum strategies to quantum isomorphisms of the corresponding graphs.

Theorem 1 ([3]). *Let $\mathcal{F}$ be a linear binary constraint system game with m constraints. Then the following are equivalent:*

- *There exists a perfect classical strategy for $\mathcal{F}$*
- *The graphs $G_{\mathcal{F}}$ and $G_{\mathcal{F}_0}$ are isomorphic*
- *$\alpha(G_{\mathcal{F}}) = m$.*

Theorem 2 ([3]). *. Let $\mathcal{F}$ be a linear binary constraint system game with m constraints. Then the following are equivalent:*

- *There exists a perfect quantum strategy for $\mathcal{F}$*
- *The graphs $G_{\mathcal{F}}$ and $G_{\mathcal{F}_0}$ are quantum isomorphic*
- *$\alpha_q(G_{\mathcal{F}}) = m$.*

When we choose $\mathcal{F}$ to be the magic square game, we deduce that $G_{\mathcal{F}}$ and $G_{\mathcal{F}_0}$ are quantum isomorphic, yet non-isomorphic graphs, as there is an optimal quantum strategy but no optimal classical strategy for $\mathcal{F}$.

3.2 Quantum Isomorphic Strongly Regular Graphs from the E_8 Root System ([20])

We continue our survey with the only known example of a strongly regular graph with a non-isomorphic, quantum isomorphic counterpart. This section summarizes the main results that lead to its discovery in [20].

Definition 7. *The E_8 root system Ψ_{E_8} consists of the $112 + 128 = 240$ vectors in $\mathbb{R}^8$, being one of the following:*

1. $|x\rangle = \pm|e_i\rangle \pm |e_j\rangle$, *for all* $1 \leq i < j \leq 8$, *or*
2. $|x\rangle = (x_1, \ldots, x_8)$ *with* $x_i \in \{\pm 1\}$ *and* $\prod_{i=1}^{8} x_i = 1$.

From this root system, we construct the orthogonality graph G_{E_8} as follows. Let $V(G_{E_8}) := \{v_x : |x\rangle \in \Psi_{E_8}, |x\rangle \equiv -|x\rangle\}$ and $E(G_{E_8}) := \{(v_x, v_y) : \langle x, y\rangle = 0\}$. This graph is strongly regular with parameters $(120, 63, 30, 36)$. Below we recall the notion of real Pauli matrices, as these will be used to describe certain automorphisms of G_{E_8}.

Definition 8. *The real (qubit) Pauli matrices are defined to be the following matrices:*

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad Y_2 = X_2 Z_2. \tag{2}$$

Recall the relations $X_2^\dagger = X_2$, $Z_2^\dagger = Z_2$, $Y_2^\dagger = -Y_2$, $X_2^2 = Y_2^2 = Z_2^2 = I_2$. Note also that from these relations it follows that X_2 and Z_2 anticommute, i.e. $X_2 Z_2 = -Z_2 X_2$. We will use these facts to show the following theorem.

Theorem 3. *The 3-qubit Pauli matrices $M := M_0 \otimes M_1 \otimes M_2$ with $M_i \in \{I_2, X_2, Y_2, Z_2\}$ induce automorphisms $\sigma_M : V(G_{E_8}) \to V(G_{E_8})$, $v_x \mapsto v_{Mx}$ which form a subgroup $L \cong \mathbb{Z}_2^6$ of $\mathrm{Aut}(G_{E_8})$.*

Proof. In order to show $\sigma_M \in \mathrm{Aut}(G_{E_8})$, we observe that the generators $I_2 \otimes I_2 \otimes X_2, I_2 \otimes I_2 \otimes Z_2$, $I_2 \otimes X_2 \otimes I_2$, $I_2 \otimes Z_2 \otimes I_2$, $X_2 \otimes I_2 \otimes I_2$, $Z_2 \otimes I_2 \otimes I_2$ are signed permutation matrices, flipping an even number of signs. As this preserves the defining properties of begin a root vector of E_8, and any element $M \in L$ is a product of these generators, we deduce that $M \in \mathrm{Aut}(G_{E_8})$. Note that due to the identification $v_x = v_{-x}$ for vectors $x \in \Psi_{E_8}$, the anti-commuting property $Z_2 X_2 = \pm X_2 Z_2$ descends to equality of automorphisms in L. As furthermore $X_2^2 = Z_2^2 = I_2$, we get $\{\sigma_{I_2}, \sigma_{X_2}, \sigma_{Z_2}, \sigma_{X_2 Z_2}\} \cong \mathbb{Z}_2^2$ for each tensor term. Moreover, since each tensor factor is independent of the other, we get $L \cong (\mathbb{Z}_2^2)^3 \cong \mathbb{Z}_2^6$.

Next we show that two vertices $v_x, v_y \in V_i$ have the same stabilizers under L.

Lemma 3. *If there exists $\sigma_M \in L$ such that $v_y = \sigma_M(v_x)$, then $\mathrm{stab}_L(v_x) = \mathrm{stab}_L(v_y)$.*

Proof. Since we have $|y\rangle = \pm M|x\rangle$ and $N|x\rangle = \pm|x\rangle \ \forall N \in \mathrm{stab}_L(v_x)$, we see that

$$N|y\rangle = \pm NM|x\rangle = \pm MN|x\rangle = \pm M|x\rangle = \pm|y\rangle,$$

since any two elements $M, N \in L$ either commute or anticommute. As vertices identify the vectors up to a sign, we conclude $\sigma_N \in \mathrm{stab}_L(v_y)$.

Moreover, the action of L partitions $V(G_{E_8})$ into 15 orbits of size 8, where each orbit comprises of a clique, or equivalently of a orthonormal basis in the E_8 root system. We denote these orbits as V_i, for $i \in \{1, \ldots, 15\}$. Furthermore, the stabilizer subgroups of each orbit, $\mathrm{stab}_L(V_i)$, are isomorphic to $\mathbb{Z}_2^3$, with the intersection of the stabilizer subgroups of two orbits being isomorphic to $\mathbb{Z}_2$.

Using these facts, it is shown that there is an induced magic unitary for G_{E_8}, coming from the action of L. For this, we choose distinguished vertices $\omega^{(i)} \in V_i$ and define projections onto the associated vectors to each orbit.

Lemma 4. *Denote $P_x := \frac{1}{||x||^2}|x\rangle\langle x|$ as the rank-one projection onto some root vector $|x\rangle \in \Psi_{E_8}$, and for root vectors $y, z \in \Psi_{E_8}$, let $M_{yz} \in L$ be the matrix under which $y \mapsto \pm z$. Define $u^{(i)}_{v_y v_z} = M_{yz} P_{\omega^{(i)}} M^*_{yz} = P_{M_{yz}\omega^{(i)}}$ and form blocks $u^{(i)} = (u^{(i)}_{v_y v_z})_{v_y, v_z \in V_i}$. Then the block diagonal matrix*

$$u = \begin{pmatrix} u^{(1)} & 0 & 0 & \ldots & 0 \\ 0 & u^{(2)} & 0 & \ldots & 0 \\ 0 & 0 & u^{(3)} & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & u^{(15)} \end{pmatrix} \tag{3}$$

is a magic unitary.

Next, the procedure is to choose distinguished vertices $\omega^{(i)} \in V_i$, and show that fixing the disjoint cliques V_i and changing the adjacencies between cliques V_i and V_j based on the adjacencies between $\omega^{(i)}$ and $\omega^{(j)}$ results in a quantum isomorphic, non-isomorphic graph G^ω, which is uniquely determined up to the isomorphism coming from different choices of $\omega^{(i)}$.

We start by defining the graph G^ω.

Definition 9. *For $\omega = \{\omega^{(1)}, \ldots, \omega^{(15)}\}$ be the set of some chosen vectors, for which $v_{\omega^{(i)}} \in V_i$ for each orbit under L. Then, the graph G^ω is constructed by setting $V(G^\omega) = V(G_{E_8})$ and switching the edges of G_{E_8} as follows:*

- *If $\langle \omega^{(i)}, \omega^{(j)} \rangle \neq 0$, we copy the edges from G_{E_8}, i.e. we define $(s,t) \in E(G^\omega) \iff (s,t) \in E(G_{E_8})$ for $s \in V_i$, $t \in V_j$.*
- *If $\langle \omega^{(i)}, \omega^{(j)} \rangle = 0$, we switch the adjacency rules from G_{E_8}, i.e. $(s,t) \in E(G^\omega) \iff (s,t) \notin E(G_{E_8})$ for $s \in V_i$, $t \in V_j$.*

Note that this procedure essentially preserves the 8-cliques making up the orbits V_i under L and switches the adjacency relations between those cliques, such that ω becomes an independent set of size 15, already showing that $G^\omega \ncong G_{E_8}$.

Moreover, even though the construction of G^ω involved making a choice on ω, it is true that two different sets ω and ω' yield graphs G^ω, $G^{\omega'}$ that are isomorphic to each other.

Finally, the magic unitary u gives a quantum isomorphism between the graphs G_{E_8} and G^ω, as is shown by the following lemma.

Lemma 5. *Let $k, s \in V_i$ and $l, t \in V_j$ with $i \neq j$.*

- *If $d(k,l) = d(s,t)$, we have $u_{ks}u_{lt} = 0$ iff $\langle \omega^{(i)}, \omega^{(j)} \rangle = 0$.*
- *If $d(k,l) \neq d(s,t)$, we have $u_{ks}u_{lt} = 0$ iff $\langle \omega^{(i)}, \omega^{(j)} \rangle \neq 0$.*

Corollary 1. *The magic unitary defined above induces a quantum isomorphism between G_{E_8} and G^ω.*

Proof. Follows, essentially by construction, from Definition 9 and Lemma 5 that certain products of u must be zero, combined with the statement from Lemma 2 that the observations from Lemma 5 imply a quantum isomorphism.

3.3 Spherical Designs

The root system E_8 is an example of a *spherical design*, which are dense and highly symmetric configurations of finitely many vectors in some Hilbert space. It is therefore reasonable to expect that other spherical designs give quantum isomorphic strongly regular graphs similar to the example of E_8. The examples that are of particular interest for this project are listed in Table 1, and were obtained from the search for particularily small spherical designs in [11]. From it, we chose to analyse the 40 vector design in $\mathbb{C}^4$ in more detail in the next section.

Table 1. Overview of spherical (t,t)-designs and their properties, extended from [11]. The entry in column *Orthogonality Graph* contains the strongly regular graph parameters.

Name	Description	Ambient	Automorphism group	Orthogonality graph
Penrose Dodecahedron	40 vector $(3,3)$-design	$\mathbb{C}^4$	$ST32 \cong \mathbb{Z}_3 \times U_4(2)$	$(40,12,2,4)$
$GQ(4,2)$	45 vector $(2,2)$-design	$\mathbb{C}^5$	$ST33 \cong \mathbb{Z}_2 \times PSU_4(2)$	$(45,12,3,3)$
Coxeter-Todd lattice Λ_6^ω	126 vector $(3,3)$-design	$\mathbb{C}^6$	$ST34 \cong \mathbb{Z}_3.\Omega_6^-.\mathbb{Z}_2$	$(126,45,12,18)$
E_6 lattice	36 vector $(2,2)$-design	$\mathbb{R}^6$	$ST35 \cong PSU_4(2).\mathbb{Z}_2$	$(36,15,6,6)$
E_7 lattice	63 vector $(2,2)$-design	$\mathbb{R}^7$	$ST36 \cong \mathbb{Z}_2 \times Sp_6(2)$	$(63,30,13,15)$
E_8 lattice	120 vector $(2,2)$-design	$\mathbb{R}^8$	$ST37 \cong \mathbb{Z}_2.O_8^+(2)$	$(120,63,30,36)$

4 A Pair of Strongly Regular Graphs from the Penrose Dodecahedron

In light of Table 1, we aim to extend the list of quantum isomorphic strongly regular graphs, by analysing the 40 vector $(3,3)$-design in $\mathbb{C}^4$ known as the Penrose

dodecahedron, which describes the 40 different states of a spin-3/2 particle and is used in its initial survey [24] to give a proof of the Kochen-Specker theorem. Motivated by the observation that its orthogonality graph Q and its clique graph Q^* share many unusual properties while not being isomorphic, we attempt to analyse this duality in order to answer whether the similarities are the shadows of an underlying quantum isomorphism. We furthermore describe how Q may be identified as the commutation graph of the 2-qutrit Pauli group, deepening our understanding of the geometry of the analysed configuration.

4.1 The Penrose Dodecahedron

The Penrose dodecahedron Φ is a set of 40 rays $|\phi_n^{(i)}\rangle, i \in \{1,\dots,4\}, n \in \{1,\dots,10\}$ represented by the vectors in $\mathbb{C}^4$ displayed in Table 2, where $\omega = e^{\frac{2\pi i}{3}}$ is the third complex root of unity. In the context of this report, a ray is defined to be the equivalence class $\{\pm\omega^d \cdot v\}$ for any $v \in \Phi$. For notational simplicity, we omit normalization factors.

Table 2. The vectors from the penrose dodecahedron.

n	$\|\phi_n^{(0)}\rangle$	$\|\phi_n^{(1)}\rangle$	$\|\phi_n^{(2)}\rangle$	$\|\phi_n^{(3)}\rangle$
0	$(1,0,0,0)$	$(0,1,0,0)$	$(0,0,1,0)$	$(0,0,0,1)$
1	$(0,1,-1,1)$	$(1,0,-1,-1)$	$(1,-1,0,1)$	$(1,1,1,0)$
2	$(0,1,-\omega,\overline{\omega})$	$(1,0,-\omega,-\overline{\omega})$	$(1,-\omega,0,\overline{\omega})$	$(1,\omega,\overline{\omega},0)$
3	$(0,1,-\overline{\omega},\omega)$	$(1,0,-\overline{\omega},-\omega)$	$(1,-\overline{\omega},0,\omega)$	$(1,\overline{\omega},\omega,0)$
4	$(0,1,-\omega,1)$	$(1,0,-1,-\omega)$	$(1,-\overline{\omega},0,\overline{\omega})$	$(1,\omega,1,0)$
5	$(0,1,-\overline{\omega},\overline{\omega})$	$(1,0,-\omega,-1)$	$(1,-1,0,\omega)$	$(1,\overline{\omega},\overline{\omega},0)$
6	$(0,1,-1,\omega)$	$(1,0,-\overline{\omega},-\overline{\omega})$	$(1,-\omega,0,1)$	$(1,1,\omega,0)$
7	$(0,1,-\overline{\omega},1)$	$(1,0,-1,-\overline{\omega})$	$(1,-\omega,0,\omega)$	$(1,\overline{\omega},1,0)$
8	$(0,1,-1,\overline{\omega})$	$(1,0,-\omega,-\omega)$	$(1,-\overline{\omega},0,1)$	$(1,1,\overline{\omega},0)$
9	$(0,1,-\omega,\omega)$	$(1,0,-\overline{\omega},-1)$	$(1,-1,0,\overline{\omega})$	$(1,\omega,\omega,0)$

4.2 The Kochen-Specker Theorem and Kochen-Specker Sets

This configuration of rays first appeared in [24] and was used to construct a proof of the Kochen-Specker theorem on non-contextuality. In short, the Kochen-Specker theorem claims the non-existence of hidden-variable theories which are compatible with the rules of quantum mechanics. The idea of its proof is to find a set of rays S, describing the possible states of some quantum system, for which a non-contextual hidden-variable labelling function $f : S \to \{0,1\}$ with $\sum_{v\in B} f(v) = 1$ for any orthonormal basis $B \subseteq S$ can not exist.

Since the original proof of the Kochen-Specker theorem, multiple such configurations were found, motivating the definition of a Kochen-Specker set.

Definition 10. *A* Kochen-Specker set S *is a configuration of rays in some Hilbert space which have no hidden-variable labelling.*

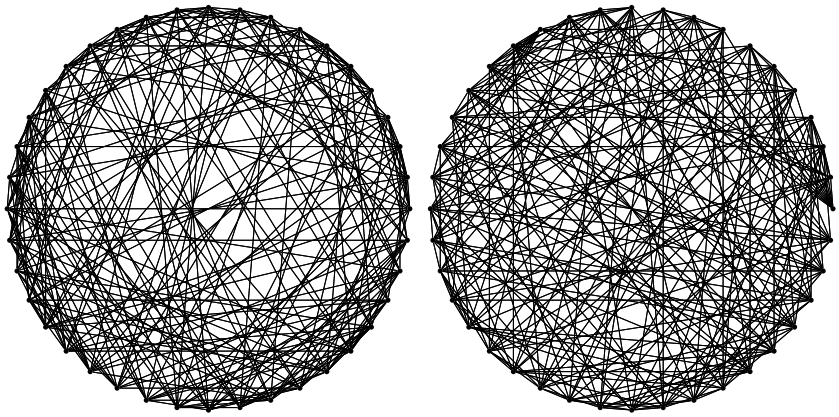


Fig. 4. Illustration of the graphs Q (left) and Q^* (right).

Note that the magic square game from Section 2.2 has essentially those properties. There is no classical labelling function f which allows the players to win with certainty.

Yet, as it turns out, there is a natural way to construct a Kochen-Specker set associated to the perfect quantum strategy magic square by considering the rays spanned by the simultaneous eigenstates of each row, giving a highly symmetric configuration in $\mathbb{R}^4$ known as the 24-cell. Respectively, the rays coming from the mutual eigenstates of each column the column form the dual of the 24-cell, see [1] and [18] for further details.

4.3 The Graphs Q and Q^*

In [24], the discussion about the non-existence of hidden-variable labelling functions included the graphs Q and Q^*, which gives the basis of the main objective for this report.

Definition 11. *We define the following graphs:*

1. *The graph Q is the orthogonality graph of Φ, i.e.*

$$V(Q) = \left\{v_i := v_{|\phi_i\rangle} \;\middle|\; |\phi_i\rangle \in \Phi\right\},\ E(Q) = \{(v_i, v_j) \iff \langle\phi_i|\phi_j\rangle = 0\}$$

2. *The graph Q^* is called the clique graph of Q, with*

$$V(Q^*) = \{v_S \mid S \subset \Phi \text{ orthonormal basis}\},$$
$$E(Q^*) = \{(v_{S_1}, v_{S_2}) \iff S_1 \cap S_2 \neq \emptyset\}.$$

In words, the vertices of Q^* correspond to the orthonormal bases of Φ, and vertices share an edge if the respective bases share a vector.

A curious duality. As noted in [24], the graphs have unusually many properties in common. As an example, the fact that $|V(Q)| = |V(Q^*)| = 40$ a priori appears to be purely coincidental. For reference, considering the graph $G^*_{E_8}$ as the clique graph of G_{E_8}, we have $|V(G_{E_8})| = 120 \neq 2025 = |V(G^*_{E_8})|$, which appear to be unrelated measures.

Moreover, the condition $S_1 \cap S_2 \neq \emptyset$ that determines whether vertices in Q^* are connected is equivalent to the condition that $S_1 \cap S_2 = \{\tilde{v}\}$ for some $\tilde{v}$, i.e. orthonormal bases in Φ share at most one ray. This gives an interesting duality between Q and Q^*, since the maximal 4-cliques in Q^* correspond to a unique vertex in Q and vice versa. Additionally, the two graphs are both strongly regular graphs with the same parameters, i.e.

$$Q, Q^* \in \mathsf{SRG}(40, 12, 2, 4).$$

Yet, as pointed out in [24], the graphs are not isomorphic, as any subgraph of Q with vertex count 10 must contain an edge, while this is not the case for Q^*. In particular, the independence number of Q is 7, while Q^* has independence number 10. We see next that these properties suggest the existence of a quantum isomorphism between Q and Q^*.

Lemma 6. *Quantum isomorphisms preserve the parameters of strongly regular graphs.*

Proof. From [6, Theorem 9.1.2], we deduce that strongly regular graphs have exactly three eigenvalues, where the first one is equal to k and the other two are called restricted eigenvalues r, s. Next, due to [6, Theorem 9.1.3], the values of r and s determine the values of n, k, λ, μ uniquely and vice versa. Since quantum isomorphic graphs are cospectral [3], meaning that their spectrum (k, r, s) is preserved, we arrive at our claim.

Counting homomorphisms from planar graphs. To strengthen the hypothesis that a potential quantum isomorphism would relate Q to Q^* and not some other graph in $\mathsf{SRG}(40, 12, 2, 4)$, we make use of a result stating that quantum isomorphic graphs admit the same number of homomorphism counts from any planar graph.

Theorem 4. ([12, Theorem 7.16]) *Let G and H be graphs, and assume $G \cong_q H$, i.e. there is a perfect quantum strategy for the (G, H)-isomorphism game. Then G and H admit the same number of homomorphisms from any planar graph.*

By iterating over all graphs in $\mathsf{SRG}(40, 12, 2, 4)$ (of which only 28 exist, see the classification in [22]), and choosing a suitable planar graph H, we may divide those 28 graphs into classes having the same homomorphism count from H. Straightforward computation shows that from those 28 graphs, only Q and Q^* have 40 4-cliques (corresponding to the vertices of the respective other graph),

while the other graphs have at most 24. Thus, taking H to be K_4, the complete graph on 4 vertices, we may directly compute the homomorphism count

$$|\mathrm{Hom}(K_4, Q)| = |\mathrm{Hom}(K_4, Q^*)| = 4! \cdot 40 = 960,$$

while the homomorphism count for any of the other graphs is at most $4! \cdot 24 = 576$. While this computation doe not prove the claim that Q and Q^* are quantum isomorphic, it shows that if any of the two graphs admit a quantum isomorphism, it has to map into the other graph of this pair.

Even more remarkably, it was shown by direct computation that there does not exist any planar graph with less than 9 vertices that distinguishes Q from Q^* via their homomorphism counts, which strengthens our conjecture that no such graph exists at all.

4.4 Q has Quantum Independence Number ≥ 10

We recall from the discussion in Section 4.2 that Q is the orthogonality graph of the Penrose dodecahedron, which forms a Kochen-Specker set. This property can be understood that the set contains k orthonormal bases, while not containing an independent set of size k. In terms of its orthogonality graph, this means that the independence number is smaller than the number of disjoint cliques.

This is indeed the case for our graph Q, as $k = 10$, yet $\alpha(Q) = 7 < 10$. From [9], we see that this already implies $\alpha_q(Q) > \alpha(Q)$. Explicitly, we give an optimal strategy for the $(Q, 10)$-independent set game which proves the bound $\alpha_q(Q) \geq 10$.

Lemma 7. *Let $\mathcal{A} = M_4(\mathbb{C})$. Then there exists a perfect strategy for the $(Q, 10)$-independent set game.*

Proof. Consider the set of projections obtained as follows.

- Choose a partition of $V(Q)$ into ten disjoint cliques C_x. This is possible as $\alpha(Q^*) = 10$.
- $P_{xu} = |v_u\rangle\langle v_u|$ if v_u is contained in clique C_x, otherwise $P_{xu} = 0$.

In order to show that it gives rise to an optimal strategy, we verify that the defining conditions from Definition 2.2 hold for our choice of projections. First, we see that $\sum_{u \in V(Q)} P_{xu} = I_4$ as we sum over the vectors contained in a fixed clique, which translates to summing over the projections of an orthonormal basis, giving I_4. Next, if $(u, v) \in E(Q)$, we know by definition that the corresponding vectors are orthogonal. Hence, even if both projections are non-zero (which is the only non-trivial case), we see that their product is nevertheless zero. Lastly, we confirm that $P_{xu}P_{yu} = 0$ for $x \neq y$. For that we make use of the fact that C_x are disjoint cliques, meaning that each vertex is a member in exactly one clique C_x. Therefore, $P_{xu} \neq 0$ for exactly one x, i.e. $P_{yu} = 0$ for all other $y \neq x$ and hence $P_{xu}P_{yu} = 0$ for $y \neq x$.

Note that this construction is essentially the one described in [9]. Furthermore, as described in [3, Lemma 5.14], it is true that quantum isomorphic graphs G, H must have the same quantum independence number, since the set of mutually non-adjacent vertices in G giving rise to a perfect quantum strategy for the (G, k)-independent set game are mapped to mutually non-adjacent vertices in H, via the quantum isomorphism between G and H.

4.5 An Interpretation of Q and Q^* in terms of Qutrit Pauli's

In this section we describe a natural way of identifying vectors of the Penrose dodecahedron in terms of qutrit Pauli matrices, answering a question in [16] and allowing to deepen our understanding of the structure of Q.

Recall that the generalized Paulis X_r and Z_r are the unique operators on $\mathbb{C}^r$ fulfilling $X_r|i\rangle = |i+1\rangle$ and $Z_r|i\rangle = \omega_r^i|i\rangle$ for all $i \in \{1, \ldots, r\}$. Throughout, we will consider the case $r = 3$, for which we have matrices

$$X := X_3 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \; Z := Z_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \bar{\omega} \end{pmatrix} \in \mathbb{C}^{3\times 3}$$

where again $\omega = e^{2\pi i/3}$ is the third complex root of unity. Recall furthermore the identities $X^2 = X^* = \bar{X}, Z^2 = Z^* = \bar{Z}$ and $X^a Z^b = \omega^{-ab} Z^b X^a$, for $a, b \in \mathbb{Z}_3$. The matrices X and Z generate the qutrit Pauli group $\mathcal{P}_3$ on $\mathbb{C}^3$, which, up to global phase, comprises of the elements

$$\left\{I, X, X^2, Z, Z^2, Y, Y^2, V, V^2\right\},$$

where $Y := XZ$, $V := XZ^2$ and I is the 3×3 identity matrix. Moreover, the 2-qutrit Pauli group $\mathcal{P}_9$ comprises of the tensor product of two elements in $\mathcal{P}_3$ and therefore has $9^2 = 81$ elements.

As noted in [16,17], the commutation relations between generalized Pauli matrices follow a nice graph-theoretical pattern. It was shown that taking the maximally commuting subsets (MCSs) of $\mathcal{P}_9 \setminus \{I \otimes I\}$ as vertices and considering two vertices to be adjacent when the corresponding MCSs share at least one element yields a strongly regular graph in $\mathsf{SRG}(40, 12, 2, 4)$. It is considered to be the dual of the so-called Pauli graph, which has $\mathcal{P}_9 \setminus \{I \otimes I\}$ as vertex set and the adjacency relation that commuting operators share an edge. However, this duality breaks down, since the Pauli graph turns out to not be strongly regular and furthermore comprises of 80 instead of 40 vertices.

This obstruction is due to the missing identification of two elements as the same vertex inside the Pauli graph. In particular, below we give a natural description of Q in terms of isotropic points in the symplectic space $\mathrm{Sp}_4(\mathbb{F}_3)$, which will immediately give the necessary identification of Pauli matrices.

The Pauli Representation of $\mathbf{Sp_4(\mathbb{F}_3)}$. There are several known constructions for Q and Q^*. For example, it is well-known from [7], that Q and Q^* are the

collinearity graphs of the two generalized quadrangles $GQ(3,3)$, which are dual to each other, meaning that points in the one correspond to lines in the other and vice versa. More interestingly, Q may also be seen as the graph containing the isotropic points inside $\mathrm{Sp}_4(\mathbb{F}_3)$, while Q^* is the isotropy graph of the $\mathrm{O}_5(\mathbb{F}_3)$ space. We will introduce both notions below.

Definition 12. *Let k be a field. A* symplectic vector space*, denoted $\mathrm{Sp}_n(k)$ or simply $\mathrm{Sp}(V)$, is a vector space $V \cong k^n$ equipped with an alternating non-degenerate bilinear form $\omega : V \times V \to k$, where the alternating property is defined as the property that $\forall u \in V$, we have $\omega(u,u) = 0$. In particular, if $\mathrm{char}(k) \neq 2$, the alternating property implies that the form is anti-symmetric, meaning that $\forall u, v \in V : \omega(u,v) = -\omega(v,u)$.*

We continue by introducing quadratic spaces, necessary to understand the structure of Q^*.

Definition 13. *[15] Let k be a field, and $V \cong k^n$ a vector space over k. A* quadratic form *over k is a function $q : V \to k$ which is defined as the evaluation of a degree-2 homogeneous polynomial in $k[x_1, \ldots, x_n]$. In particular,*

$$q(\lambda v) = \lambda^2 q(v)$$

for any $\lambda \in k$. Then, the n-dimensional quadratic space over k*, denoted as $\mathrm{O}_n(k)$ is defined as the vector space V equipped with some quadratic form q.*

Moreover, two vectors $v_1, v_2 \in \mathrm{O}_n(k)$ are said to be mutually isotropic *whenever $q(v_1, v_2) = 0$. Similarily, a vector v is isotropic iff $q(v,v) = 0$.*

The reason for this notation comes from the fact that the closely related *orthogonal group* $O_n(k) \subseteq GL_n(k)$ over the form q is the group of matrices $M \in GL_n(k)$ that leave the underlying quadratic space invariant, i.e. $q(Mv) = q(v)$.

Next we introduce the notion of a point in either of these spaces.

Definition 14. *Let V be either a symplectic or a quadratic space. A* point *in V is a 1-dimensional subspace $\langle v \rangle \subseteq V$ spanned by some vector $v \in V$. Throughout, we often identify points with some representative vector spanning this subspace.*

Interestingly, in [7] it is mentioned that the strongly regular graphs Q and Q^* from above can be interpreted as the isotropy graphs of the points in $\mathrm{Sp}_4(\mathbb{F}_3)$ and $\mathrm{O}_5(\mathbb{F}_3)$, respectively. Furthermore, the adjacency relation is defined by isotropy of the corresponding two points in the respective space. This may be reformulated further, as there is a representation $\mathrm{Sp}_4(\mathbb{F}_3) \to GL_9(\mathbb{C})$ given by the 2-qutrit Pauli group $\mathcal{P}_9$.

Lemma 8. *Let $\mathrm{Sp}_4(\mathbb{F}_3)$ be the vector space $\mathbb{F}_3^4$ equipped with the symplectic form $\omega(u,v) := u_1v_2 - v_1u_2 + u_3v_4 - v_3u_4$. Then there is a canonical map from the isotropic points in $\mathrm{Sp}_4(\mathbb{F}_3)$ to $\mathcal{P}_9$, defined by the rule*

$$\langle v \rangle := (v_1, v_2, v_3, v_4) \;\mapsto\; X^{v_1}Z^{v_2} \otimes X^{v_3}Z^{v_4} =: \rho_v \tag{4}$$

Furthermore, for $u, v \in \mathrm{Sp}_4(\mathbb{F}_3)$ we have that u and v are mutually isotropic iff ρ_u and ρ_v commute.

Proof. First, note that due to the properties of a symplectic form, we have $\omega(v, v) = 0$ for any vector $v \in \mathrm{Sp}_4(\mathbb{F}_3)$, i.e. any point is trivially isotropic.

Next, for any 2-qutrit Pauli matrices ρ_u, ρ_v we observe

$$\begin{aligned}
\rho_u\rho_v &= (X^{u_1}Z^{u_2} \otimes X^{u_3}Z^{u_4})(X^{v_1}Z^{v_2} \otimes X^{v_3}Z^{v_4}) \\
&= X^{u_1}Z^{u_2}X^{v_1}Z^{v_2} \otimes X^{v_3}Z^{v_4}X^{u_3}Z^{u_4} \\
&= \omega^{v_1u_2}X^{v_1}X^{u_1}Z^{v_2}Z^{u_2} \otimes \omega^{u_3v_4}X^{v_3}X^{u_3}Z^{v_4}Z^{u_4} \\
&= \omega^{v_1u_2}\omega^{-u_1v_2}X^{v_1}Z^{v_2}X^{u_1}Z^{u_2} \otimes \omega^{u_3v_4}\omega^{-v_3u_4}X^{v_3}Z^{v_4}X^{u_3}Z^{u_4} \\
&= \omega^{v_1u_2-u_1v_2+u_3v_4-v_3u_4}(X^{v_1}Z^{v_2} \otimes X^{v_3}Z^{v_4})(X^{u_1}Z^{u_2} \otimes X^{u_3}Z^{u_4}) \\
&= \omega^{v_1u_2-u_1v_2+u_3v_4-v_3u_4}\rho_v\rho_u.
\end{aligned}$$

Thus, ρ_u and ρ_v commute iff $v_1u_2 - u_1v_2 + u_3v_4 - v_3u_4 = 0$, which is exactly the condition that $\omega(u, v) = 0$, i.e. that u and v are mutually isotropic.

Furthermore, recall that the symplectic form ω is bilinear, i.e. $\omega(\lambda u, v) = \lambda\omega(u, v)$ for $\lambda \in \mathbb{F}_3^\times$, and similarily for the other argument. Therefore, we see that vectors u, v being mutually isotropic extends to their spanned 1-dimensional subspaces, i.e. the associated points, being isotropic. We must therefore identify those Pauli matrices whose symplectic vectors describe the same point. Since $\mathbb{F}_3^\times = \{1, 2\}$, we shall identify each ρ_v with $\rho_{2v} = \rho_v^2 = \rho_v^\dagger$, meaning that Pauli operators and their *inverses* form one point in $\mathrm{Sp}_4(\mathbb{F}_3)$ and thus form the same vertex in Q.

Strikingly, there exists a representation of $\mathrm{O}_5(\mathbb{F}_3)$ that is equivalent to the one described above. It is a consequence of the natural map

$$\mathrm{Sp}_4(\mathbb{F}_3) \to \mathrm{O}_5(\mathbb{F}_3)$$

induced by the second exterior power, as described in [5]. In essence, this map comes from the isomorphism $\mathrm{O}_5(\mathbb{F}_3) \cong \wedge^2\mathbb{F}_3^4/\langle\omega\rangle$, where $\omega := e_1 \wedge f_1 + e_2 \wedge f_2$ is the induced bivector by the symplectic form. This suggests identifying vertices in Q^* as bivectors over $\mathrm{Sp}_4(\mathbb{F}_3)$.

Remark 4. While the identification of $V(Q)$ with 2-qutrit Paulis is rather natural, it is yet unclear how to canonically map points in $\mathrm{Sp}_4(\mathbb{F}_3)$ to the vectors inside the Penrose dodecahedron.

5 Discussion

We have seen that the graphs arising from the Penrose dodecahedron share many unusual properties, such as both being strongly regular with the same parameters, having compatible quantum independence numbers, and being the clique graph of each other, which all hint to the fact that they might be quantum isomorphic. Moreover, no property was found that definitely rules out this conjectured quantum isomorphism.

Below we attempt to apply the method from [20] in order to directly construct this quantum isomorphism, and show that unfortunately, there are several structural hurdles that complicates the search for a suitable magic unitary.

5.1 Applying the Approach from [20]

We firstly observe that Q^* has independence number 10 and, by straightforward computation, see that there are a total of 2880 maximally independent sets, which translates to the fact that there are 2880 possible partitions of $V(Q)$ into disjoint 4-cliques. However, no subgroup of $\mathrm{Aut}(Q)$ exists that induces 10 orbits comprising of such cliques, unlike was found for G_{E_8} in [20].

Secondly, it is impossible to apply the construction of G^ω in [20] to Q in order to receive a graph isomorphic to Q^*. This is due to the following observations. Firstly, the induced subgraph of two disjoint cliques in Q do not share the property of the analogous subgraphs of G_{E_8} that for each vertex inside clique V_i, there are exactly as many neighbours in V_j as there are non-neighbours. This allowed to switch the adjacency relations while preserving the edge count. However, for both Q and Q^*, each vertex in one clique V_i has exactly one neighbour and three non-neighbours in each V_j, meaning that applying this form of switching changes the edge count. However, this contradicts the fact that Q and Q^*, which have the same edge count, are related via this form of switching.

More importantly, the duality between Q and Q^* that vertices in one graph are 4-cliques in the other forbids us to fix a partition of disjoint 4-cliques, since any graph resulting from this procedure would still comprise of 10 disjoint 4-cliques. Assuming that such a graph is isomorphic to Q^* implies that Q has independence number 10, which is a contradiction. Therefore, the magic unitary corresponding to the potential quantum isomorphism cannot be a block diagonal matrix with 10 blocks of size 4 by 4.

Moreover, it is unresolved whether either of the graphs has quantum symmetry, as it was shown by exhaustive enumeration that there are no disjoint automorphisms in $\mathrm{Aut}(Q)$, which would imply the existence of a quantum automorphism, as proved in [19].

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